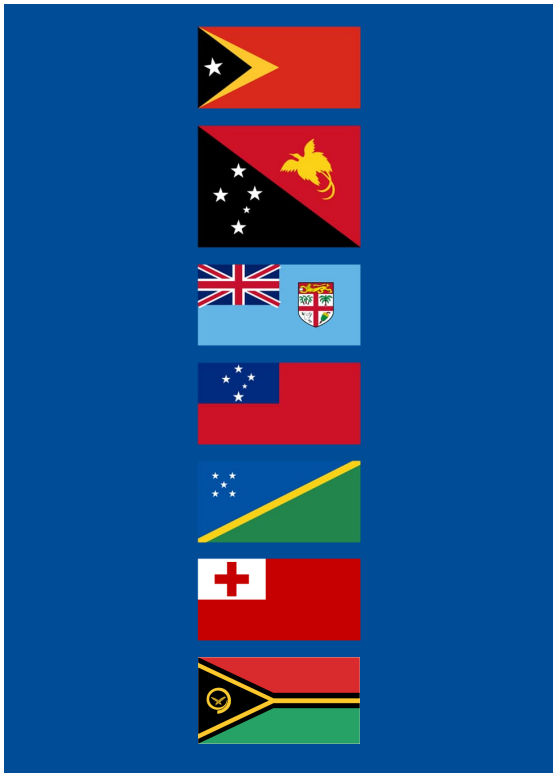


L-3: Introduction to High-Frequency Indicators

OT26.07 Cohort Training in Forecasting and Nowcasting
September 21 – October 2, 2026

Mr. Paul Cashin
Director
IMF – STI

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Outline of Lecture

- Data Sources
  - Overview of Available Data Sources
  - Examples of Non-traditional Data Sources
  - Using Google Places API and Google Trends data for HFIs
- Understanding Data Cleaning and Aggregation Processes
  - Data Cleaning
  - Data Processing
  - Data Transformations

Data Sources

Overview of Available Data Sources

- **Traditional Data Sources Used to Compile GDP**
  - Household Surveys
  - Business Surveys
  - Government Finance Statistics
  - External Sector Statistics
  - Monetary and Financial Statistics
  - Price Statistics
  - Administrative data
  - Other – non-traditional data sources



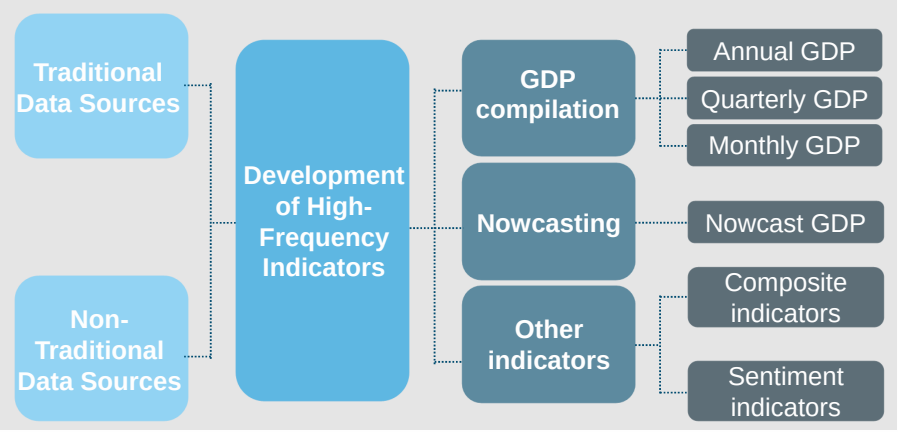
Overview of Available Data Sources

- **Traditional Data Sources Used to Compile GDP – Main Characteristics**

Characteristic	Census	Administrative Data	Sample Survey
Coverage	Complete	High, to near complete	Partial, requires weighting up
Detail	Very high; can cover different statistical needs	From low to high; can be inconsistent with SNA	As high as needed, but can be limited by respondent burden
Conceptual consistency	Can vary from SNA concepts	Some adaptations are usually required	Usually good but can require updating to latest SNA standards
Timeliness	5- or 10-yearly; economic shocks in reference periods can undermine results	Usually annual or sub-annual; can cope with and help observe economic shocks	Usually annual or sub-annual; can cope with and help observe economic shocks
Financial cost	Very high	Very low	Medium to low
Staffing needs	Very high; requires additional temporary staff	Low; may require some dedicated staff	Medium to low; still usually requires dedicated teams
Sampling error	None as covers population	None, unless a sample is taken from full set of returns	Can be high; especially if response rates are low or variable
Non-sampling error	Can be high; not possible to fix all design and processing flaws ex post	Can be specific compliance issues	Can be reduced over time through improved survey design and processing

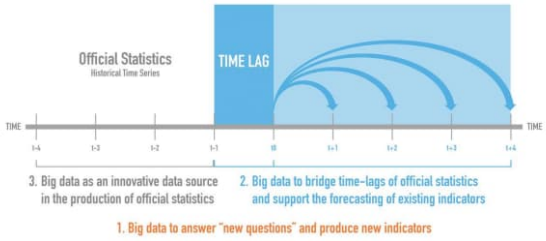
Overview of Available Data Sources

- Compilation of GDP and other macroeconomic statistics traditionally relies on data sources that are available with a time lag.
  - With **digitalization** and increasing **availability of administrative data**, a new range of non-traditional data sources is emerging and can potentially be used to improve estimates of GDP, as input for nowcasting exercises, and for the development of high frequency economic indicators.



Overview of Available Data Sources

- **Potential uses of Big Data**
  - Answer new questions and produce new indicators
  - Bridge the time-lags of official statistics and support nowcasting and forecasting exercises
  - Innovative data source to produce official statistics



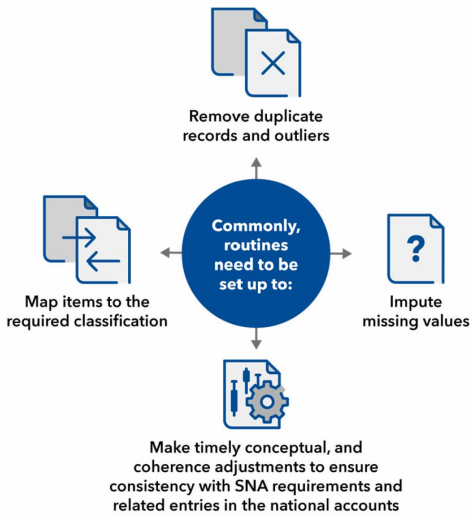
Available at: https://www.imf.org/en/Publications/Staff-Discussion-Notes/Issues/2017/09/13/Big-Data-Potential-Challenges-and-Statistical-Implications-45106

- **Specific uses of Big Data**

Data Origin	Data Source and Technique	Potential Indicators
Internet searches on search engines	**Google** (trends and search data)	• Nowcast GDP • Nowcast employment • Consumer sentiment • Car and property sales
Data produced by commercial businesses	**Swift data** on transactions, quantities, and financial market prices	• Export import indicators • Global financial flows
	Web scraping business websites	• Enterprise profile
Data from sensors	**GPS positioning** and tracking data	• Travel services imports and exports
	Traffic or road sensors	• Proxy of economic growth
	Satellite imagery (research and mapping of weather and climate data)	• Improved geographical localization of statistical units and assets
	Smart meters (energy consumption measures)	• Household consumption • Electricity supply and consumption

Overview of Available Data Sources

- **Non-traditional data sources present increased timeliness, frequency, and granularity, but some challenges need to be addressed:**
  - Securing **data access** on a regular basis with consistent coverage, available variables, and frequency;
  - Ensure enough **coverage** of economic activity and time series that are long enough;
  - **Potential conceptual misalignment** between the Big Data source and the target statistic being produced, as the Big Data source was in general not developed with the objective of producing statistics.

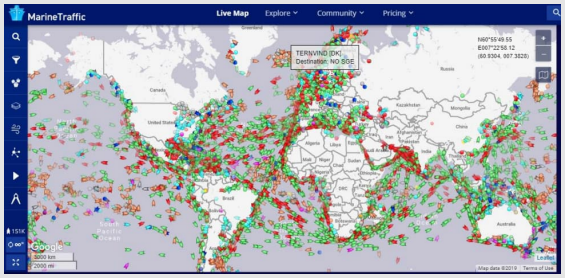


Examples of Non-traditional Data Sources

- **Automatic Identification System (AIS) Data**
  - Information include location, speed, and status of vessels (e.g., weekly port calls and trade volume).
  - Possible use to nowcast trade.

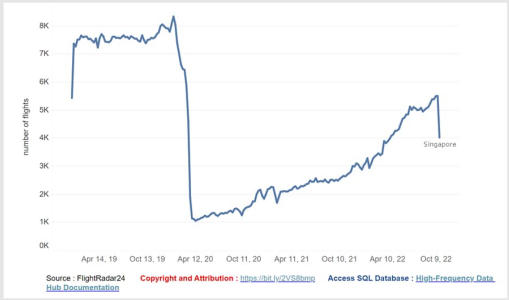
- **Flight Data**
  - Possible use in air travel estimates, estimates for the tourism activity.

A Snapshot of Global Vessel Traffic Based AIS Data



Source: https://www.imf.org/~/media/Files/Publications/WP/2019/wpiea2019275-print-pdf.ashx, based on MarineTraffic Data.
Note: Different types of vessels are shown in different colors.

Singapore: Domestic and International Flights

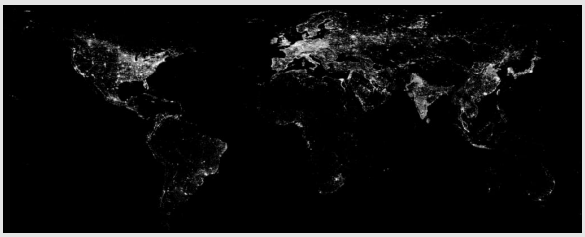


Source: FlightRadar24.

Examples of Non-traditional Data Sources

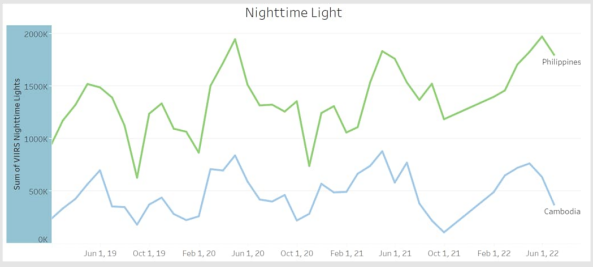
- **Earth Observation Data**
  - Using satellite imagery data to improve official statistics on a wide range of topics including agriculture, climate, business activity, and transport.
  - Pilot projects include crop density, agricultural statistics, land cover and use statistics, urban-rural systems, climate data, and crude oil inventory.

Map of Nighttime Lights in 2010



Source: https://www.imf.org/en/Publications/WP/Issues/2019/04/09/Illuminating-Economic-Growth-46670.

Cambodia and Philippines: Example of Indicator based on Nighttime Data



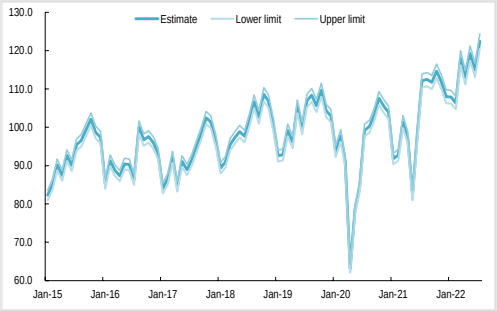
Source: Earth Observation Group, Payne Institute for Public Policy and IMF, Beyer, Hu, and Yao, *Measuring Quarterly Economic Growth from Outer Space*, 2022.

Examples of Non-traditional Data Sources

- **Electricity Consumption Data**
  - Possible use to estimate activity of economic sectors as manufacturing industry, or as input to nowcast GDP.

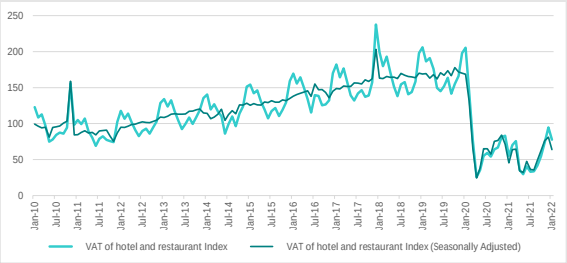
- **Tax Data**
  - Value Added Tax (VAT) and other tax data capture both turnover and expenditure
  - Possible use to track trade, hotels and restaurants, and other activities

Colombia: Early Estimation Indicator for Manufacturing Industry and 95 Percent Confidence Intervals



Source: IMF, based on DANE data (as of September 12, 2022), available at Estadísticas experimentales (dane.gov.co).

Bank of Thailand: Private Consumption Index and Components



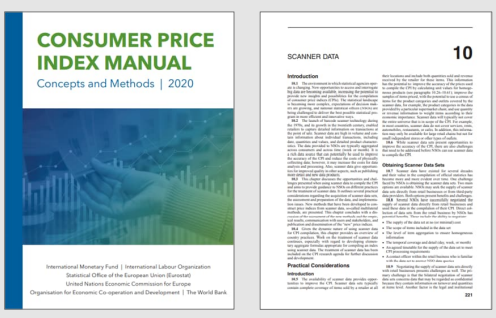
Source: IMF staff, based on Bank of Thailand data available on the website on March 10, 2022 (https://www.bot.or.th/English/Statistics/Indicators/Pages/default.aspx)

Examples of Non-traditional Data Sources

- **Scanner Data**
  - Use of scanner data from supermarket chains and other retailers, as well as online prices obtained from web scraping, to compile price indices

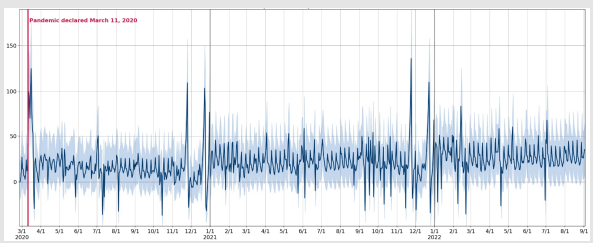
- **Credit Card Data**
  - Possible use to understand changes in consumption patterns and estimate consumption expenditure

2020 CPI Manual: Chapter on Scanner Data



Source: https://www.imf.org/-/media/Files/Data/CPI/cpi-manual-concepts-and-methods.ashx.

United States: Spending on Food and Beverage Stores – An Event Study Based on Payment Card Transactions

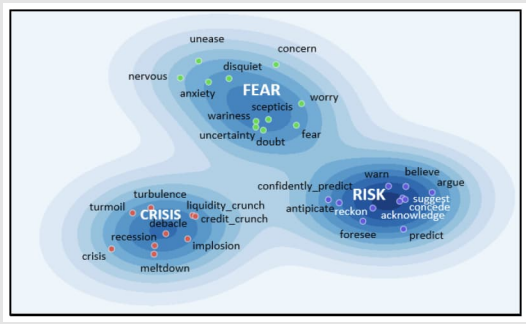


Source: US Bureau of Economic Analysis, available at https://www.bea.gov/recovery/estimates-from-payment-card-transactions (as of September 13, 2022).
Note: Chart shows the difference from the typical level of spending without COVID-19-related changes in the economy. The typical level corresponds to a value of 0. The shaded area represents 95 percent confidence interval bands.

Examples of Non-traditional Data Sources

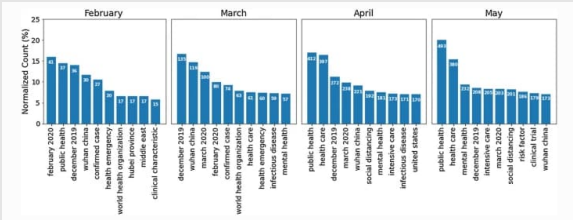
- **Text Data**
  - Natural Language Processing
  - News-based sentiment Indicators: construct sentiment indices for 20 countries from 1980 to 2019, relying on computational text analysis (Huang et al, 2019).
  - Understanding the temporal evolution of COVID-19 research (Ebadi et al, 2021).
  - Identifying the latent topics and analyzing the temporal evolution of the extracted research themes, publications similarity, and sentiments

Semantic Clustering enables to identify sentiment
(Huang et al, 2019)



Source: IMF Working Paper, News-Based Sentiment Indicators. Available at https://www.imf.org/en/publications/wp/issues/2019/12/06/news-based-sentiment-indicators-48740

Frequent keyphrase in COVID-19 publication
(Ebadi et al, 2021)



The top-10 most frequent keyphrases in COVID-19 publications. The numbers on the bars represent the exact frequency of the respective keyphrase

Source: Ebadi, A., Xi, P., Tremblay, S. et al. Understanding the temporal evolution of COVID-19 research through machine learning and natural language processing. Scientometrics 126, 725–739 (2021). https://doi.org/10.1007/s11192-020-03744-7

Using Google Earth Engine for Satellite Imagery

- **Google Earth Engine**
  - A platform designed for the scientific analysis and visualization of geospatial data.
  - It stores and manages a vast public archive of satellite imagery, including historical Earth observations spanning more than four decades.
  - New images are added daily, making the data available for large-scale, global analysis.
  - You can analyze forest and water coverage, land use change, among many other possible analyses.
- **"Poverty from Space" (Babenko et al., 2017)**
  - Uses high-resolution satellite imagery and Convolutional Neural Networks (CNNs) to estimate asset-based poverty across Sri Lanka.
  - Simple linear models explain 60–61% of the variation in poverty and average log per capita consumption.
  - Key predictors include building density, built-up area, and shadow coverage—effective in both urban and rural settings.

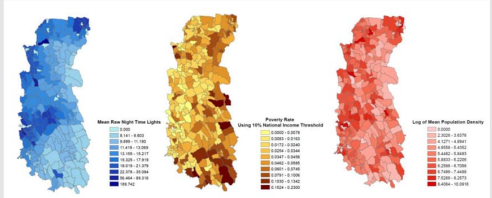
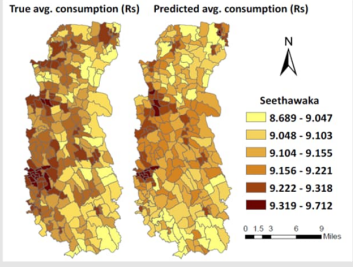


Figure 2: Comparison of Mean Night Time Lights (NTL), Poverty Rate, and Mean Population Density, Seethawaka, Sri Lanka



Source: Babenko, B., Hersh, J. S., Newhouse, D. L., Ramakrishnan, R., & Swartz, T. (2017). *Poverty from space: Using high-resolution satellite imagery for estimating economic well-being*. World Bank Policy Research Working Paper No. 8284. https://doi.org/10.1596/1813-9450-8284

Using Google Places API/ Trends Data for HFIs

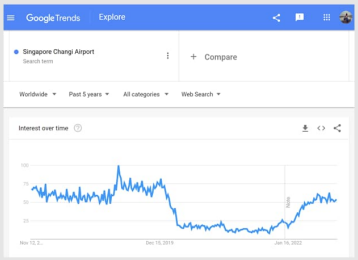
- **Google Places API**
  - The Google Places API allows users to extract information about Places from the Google Maps Platform.
  - Users can extract 23 fields of information for each Place, including name, address, places ID, business status, latitude/longitude, reviews, pricing level, and hours of operations.

- **Google Trends Data**
  - Google Trends are a measure of interest in a topic relative to all other topics over time. A topic can be anything from a person or event to a business or specific product.
  - Numbers represent search interest relative to the highest point on the chart for the given region and time. A value of 100 is the peak popularity for the term. A value of 50 means that the term is half as popular. A score of 0 means there was not enough data for this term.
  - When topics relate to a business, industry, or product the trend could be indicative, at least to some extent, of economic activity.

Example of Google Places API Data



Example of Google Trends Data



Using Google Places API/ Trends Data for HFIs - IMF

- **Method: (A) Opening Status Indicators**
  - Places are given the status of “Open,” “Temporarily Closed” or “Permanently Closed.” This information can be used to produce several useful business dynamic indicators.

Operational Indicator

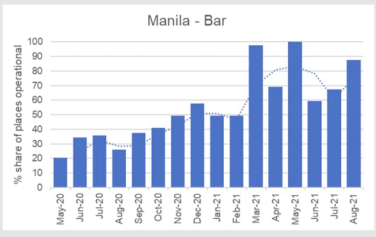
- Represents the share of Places in each geographic region that are operational at a given point in time weighted by the number of reviews.
- Weighting by reviews is intended to capture the impact of the size of the business, in which businesses with more reviews will have a larger impact on the movement in the indicator.

Example of Business Status by Week

Week 1 – Initial Status: Places A, B, C, and D operational				
Business	Reviews	Share of Reviews	Business Status	Operational Indicator
A	1000	40	Operational	
B	500	20	Operational	
C	500	20	Operational	
D	100	4	Operational	
E	400	16	Temporarily Closed	
	2500	100		84
Week 2 – Place F is temporarily closed				
Business	Reviews	Share of Reviews	Business Status	Operational Indicator
F	1000	37	Temporarily Closed	
G	700	26	Operational	
H	500	19	Operational	
I	100	4	Operational	
J	400	14	Operational	
	2700	100		63
Week 3 – All places are operational				
Business	Reviews	Share of Reviews	Business Status	Operational Indicator
F	1000	33	Operational	
K	700	24	Operational	
L	500	17	Operational	
M	400	13	Operational	
J	400	13	Operational	
	3000	100		100

Source: www.elibrary.imf.org/view/journals/001/2021/295/001.2021.issue-295-en.xml.

Example of Operational Indicator: Manila - Bars



Source: www.elibrary.imf.org/view/journals/001/2021/295/001.2021.issue-295-en.xml.

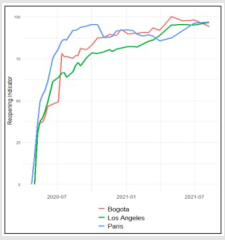
Using Google Places API/ Trends Data for HFIs - IMF

- **Method: (A) Opening Status Indicators**
  - Places are given the status of “Open,” “Temporarily Closed” or “Permanently Closed.” This information can be used to produce several useful business dynamic indicators.

Business Re-opening Indicator

- Used to track the path and pace at which businesses that are temporarily closed in a region re-open, being of particular interest to assess the impact of government regulations during the COVID-19 pandemic.
- The indicator starts with the selection of a baseline cohort of places that are temporarily closed. Each week the status of each of these Places is examined to see if they have opened or remain temporarily closed.
- The indicator represents the share of Places in each geographic region that are operational at a given point in time weighted by the number of reviews.

Business Re-opening Indicator for Selected City Centers



city	sample size	baseline	24-Apr-20	24-May-20	26-Jul-20	26-Aug-20	3-Nov-20	30-Jan-21	31-Mar-21	1-May-21	1-Jul-21	10-Aug-21
Atlanta	503	2-May-20		59	82.9	96	97	96.8	97.6	98.6	98.8	99
Bogota	339	2-May-20		37.2	76.4	76.7	87.6	90	92	100	97.9	94.1
Casablanca	209	17-May-20		9.1	32.5	40.7	47.4	54.5	60.3	93.8	65.1	67.9
Istanbul	566	24-Apr-20	0	41.7	64.7	78.3	83.6	83.9	88.5	94.9	88.7	91.5
Lagos	180	24-Apr-20	0	25	38.3	38.9	53.9	58.3	62.8	98.3	76.7	76.1
London	842	24-Apr-20	0	53.4	84.8	93.1	97.1	80.8	79	89.7	97.9	98.5
Los Angeles	1,001	2-May-20		40	64	72.8	79.1	82	88.8	95	95	96.6
Madrid	1,437	24-Apr-20	0	44.6	75.9	87.3	92.3	92.9	94.7	98.3	95.8	95.9
Manila	2,750	24-Apr-20	0	41	70.1	79.6	84.6	88.1	89.8	96.6	92.1	92.8
Milan	936	24-May-20		0	59.1	77.1	84	81.9	86.3	95.8	92	90.5
Mumbai	2,939	24-Apr-20	0	45.6	66	72.8	85.7	92.8	94.2	97.2	93.4	93.9
New York	1,278	24-Apr-20	0	47.4	75.7	84.7	92.4	92.3	94.3	96.9	97.2	97
Paris	1,645	24-Apr-20	0	53.5	86	92.5	87.7	89.7	85.6	87.4	96.2	96.9
Rome	1,343	17-May-20		25.4	64.6	82.7	88.2	88.3	89.4	97.2	93.8	94.6

Source: www.elibrary.imf.org/view/journals/001/2021/295/001.2021.issue-295-en.xml.

Using Google Places API/ Trends Data for HFIs - IMF

- **Method: (B) Business Activity Indicators**
  - Google Trends and Reviews are used as a proxy to business activity, assuming that there is a relationship between changes in interest in a topic(s) and changes in business activity.

Google Reviews as an Indicator of Business Activity

- The Google Places API permits users to extract the number of reviews posted for a given Place and the average rating (ratings range from 1 - poor to 5 - excellent). It is assumed higher change in ratings are correlated with higher economic activity.
- The rating was used to adjust the number of reviews such that a Place with 100 poorly rated reviews would have a lower weight than a Place with 100 highly rated reviews.

Example of Review Activity Indicator: Manila – Change in Restaurant Review



Source: www.elibrary.imf.org/view/journals/001/2021/295/001.2021.issue-295-en.xml.
Note: The dotted lines in the graphs are three-month moving averages.

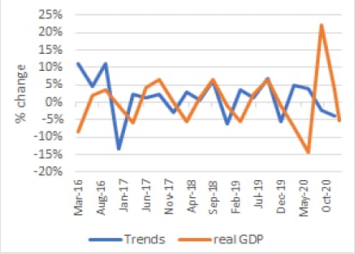
Using Google Places API/ Trends Data for HFIs - IMF

- **Method: (B) Business Activity Indicators**
  - Google Trends and Reviews are used as a proxy to business activity, assuming that there is a relationship between changes in interest in a topic(s) and changes in business activity.

Google Trends as an Indicator of Activity

- Given the infinite number of possible search terms, Google has developed an algorithm to aggregate searches into "trend" categories (e.g., category of "Consumer Electronics").
- The Google Trends categories are then mapped to the *International Standard Industrial Classification of All Economic Activities* (ISIC), to create indicators of business activity.

Example of Business Activity Indicator: Brazil - Wholesale and retail trade; Repair of motor vehicles and motorcycles; Transportation and storage; Accommodation and food service activities



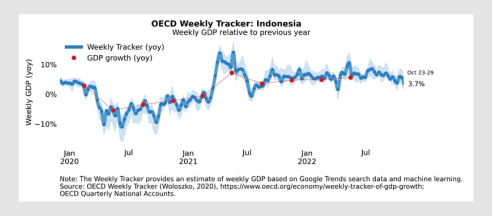
Source: www.elibrary.imf.org/view/journals/001/2021/295/001.2021.issue-295-en.xml.
Note: Indicator refers to ISIC Rev. 4, Sections G, H, and I.

Using Google Places API/ Trends Data for HFIs - OECD

- The **OECD Weekly Tracker of GDP growth** provides a real-time high-frequency indicator of economic activity using machine learning and Google Trends data.
  - Available for 46 OECD and G20 economies.
  - Applies a machine learning model to a panel of Google Trends data and aggregates information about search behavior related to consumption, labor markets, housing, trade, industrial activity and economic uncertainty.
- There are three series of the Weekly Tracker:
  - **Tracker (yoy):** estimates of weekly GDP relative to the same week in the previous year. Covers the period from early 2020 to today.
  - **Tracker (yo2y):** estimates of weekly GDP relative to the same week in the two years before (the 104-week difference). Covers the period from early 2020 to today.
  - **Tracker (counterfactual):** estimate of the percent difference between weekly GDP and the pre-crisis GDP trend (as proxied by OECD forecasts made in November 2019). Available until the end of 2021.

Each series has its own 95% confidence intervals (lower and higher bands).

The OECD Weekly Tracker for Indonesia



Source: https://www.oecd.org/economy/weekly-tracker-of-gdp-growth/.

 Source: https://www.oecd.org/economy/weekly-tracker-of-gdp-growth/.

Understanding Data Cleaning and Aggregation Processes

Data cleaning and aggregation processes

Non-traditional data sources are generally collected, stored, processed and disseminated by institutions, either private or public, which are not subject to the **quality assurance framework** imposed to the producers of official statistics.

This means that

- data can include **much more anomalies** than traditional statistics.
- the same nature of this data, related to human activities or to automatic detection of phenomena, is associated to a high number of data irregularity due to errors, mismeasurement etc.

Data Processing

Data in the real world are dirty

- **Incomplete:** presence of missing values or lack of certain attributes of interest. Incomplete data may come from: “Not applicable” data value when collected, Human/hardware/software problems
- **Noisy:** containing errors or outliers, e.g., Salary=“-10” (an error).
  Noisy data (incorrect values) may come from: faulty data collection instruments, human or computer error at data entry, errors in data transmission
- **Inconsistent:** containing discrepancies in codes or names, presence of transitory or permanent shifts, inconsistency of pattern over the time.
  Inconsistent data may come from: different data sources, functional dependency violation (e.g., modify some linked data)
- **Intentional:** (e.g., disguised missing data) which unknown, inapplicable, or otherwise non specified responses are encoded as valid data values

Data Processing

Major Tasks in Data Preprocessing

Data Cleaning	Data Transformation and Discretization
- Fill in missing values - Smooth noisy data - Identify or remove outliers - Resolve inconsistencies	- Aggregation - Normalization - Indexing - Growth rates

Data Cleaning – Missing Data

Characterization of missing values

- **Missing completely at random (MCAR)** → the probability of being missing is the same for all cases.
  E.g., *a questionnaire might be lost in the post.*
- **Missing at random (MAR)** → the probability of being missing is the same only within groups defined by the observed data.
  E.g., job title is missing when employment status is *Unemployed.*
- **Not missing at random (NMAR)** → the probability of being missing varies for reasons that are unknown to us.
  E.g., *in public opinion research MNAR occurs if those with weaker opinions respond less often*.

Depending on the type of missing value, some of the handling methods will be suitable or not.

Data Cleaning – Missing Data

Strategies for missing values handling

- **Discarding examples with missing values:** Simplest approach. Only practical if there are few examples with missing values. Otherwise, it can introduce bias.
- **Imputation based on most common value:**
  - If the missing value is continuous: Replace it with the mean value
  - If the missing value is discrete: Replace it with the most frequent value
  - Simple and fast to compute: Assumes that the data presents a normal distribution
- **Imputation methods:** Assign a value to the missing one, based on the rest of the dataset. As they extract a model from the dataset to perform the imputation, they are suitable for MCAR and, to a lesser extent, MAR types of missing values. Not suitable for NMAR type of missing data.

Data Cleaning – Outliers

- **Outliers** are data objects with characteristics that are considerably different than most of the other data objects in the dataset.
  - It is recommended to correct as much as possible such anomalies in order to obtain more regular time series with a minimum of anomalies.

- **Strategies for handling outliers**
  - **Z-scores:** A common approach is to estimate "z" normal scores (differences between each value and the mean divided by the standard deviation). A z-score describes the position of a raw score in terms of its distance from the mean when measured in standard deviation units. It allows to calculate the probability of a score occurring within a standard normal distribution.
  - **Smoothing:** can be used to remove noisy points from a time series. (See *Strategies for handling seasonality*).

Data Cleaning – Seasonality

- Since most big data are collected and recorded at high or even very-high frequency, it is likely that they are characterized by **seasonal or other very short run periodicities**. It is also possible that they can be subject to **calendar effects**.

- **Seasonality** in a time series might be caused by a variety of factors, such as weather, fixed holidays, trading times, etc. In addition, **calendar effects** can also matter, such as those generated by moving holidays (e.g., Easter). As seasonal patterns and calendar effects can bias or hide the relationship between the big data indicators and the target variable of the forecasting exercise, it is preferable to remove them and work with adjusted indicators.

- In case of the presence of such effects, it has to be **decided if the new indicator should be presented** raw, seasonally or seasonally and calendar adjusted, or both.

Data Cleaning – Seasonality

Strategies for handling seasonality

- When dealing with nowcasting exercises, high frequency indicators should be subject to a similar filtering process as the target variable.
- **Smoothing**: remove noise from data. Smoothing is usually done to help us better see patterns, trends for example, in time series. Generally smooth out the irregular roughness to see a clearer signal. For seasonal data, we might smooth out the seasonality so that we can identify the trend. Smoothing doesn't provide us with a model, but it can be a good first step in describing various components of the series.
  - **Moving Averages:** at each point in time we determine (possibly weighted) averages of observed values that surround a particular time.

Data Cleaning – Seasonality

- **Decomposition:** separate trend and seasonal component from the time series.
  - **Simple:** Decompose a time series into seasonal, trend and irregular components. E.g., *stl* package in R identifies the seasonal component using *loess* smoothing and extracts the trend.
    → You must know if you're dealing with an additive or multiplicative model.
    → Assumes series has already been corrected from outliers and calendar effects.
  - **Advanced:** seasonal adjustment packages for an automatic detection facility to detect and correct for outliers and calendar effects followed by simple decomposition. These statistical algorithms have usually be designed specifically for monthly and quarterly time series and are applied before the decomposition of the time series in trend-cycle, seasonal component and irregular.
    → tramoseats: Seasonal Adjustment with TRAMO-SEATS method
    → X-13 ARIMA

Data Transformations

Useful transformations:

- **Aggregation**: summarization, Counting, Volumes
  - High frequency data refers to time-series data collected at an extremely fine scale, both temporal and geographical. This can provide relevant information on units, such as regions or sectors, not fully covered by traditional coincident and leading indicators.
- **Normalization**: scaled to fall within a small, specified range
- **Indexing**: timeseries is indexed to a common starting point. Indexed data are handy because they allow an observer to quickly determine rates of growth by looking at a chart's vertical axis. They also allow for comparison of variables with different magnitudes.
- **Growth rates**: Growth rates refer to the percentage change of a specific variable within a specific time period, generally the same period in previous year.

Thank you!